CL-PSDD: Contrastive Learning for Adaptive Generalized Pavement Surface Distress Detection

Ruchan Dong, Jinwei Xia, Jin Zhao, and Lei Hong

***Abstract*— The accurate detection of pavement surface distress is crucial for timely maintenance and prevention of traffic accidents. Traditional deep-learning techniques, while effective, rely heavily on supervised learning, necessitating extensive and costly annotated datasets. This paper presents CL-PSDD(Contrastive Learning for Adaptive Generalized Pavement Surface Distress Detection), a novel unsupervised contrastive learning approach that overcomes these limitations by capturing robust feature representations of pavement surface distress across various conditions without manual labeling. We introduce a synergistic methodology that combines SwAV for effective unsupervised learning and YOLO for efficient object detection, enabling the model to learn intricate feature representations of pavement surface distress. Furthermore, we propose RE-RPN-K(Road Enhanced Region Proposal Network in K), a specialized region proposal network that distills precise road surface information, and integrates a transformer-based DETR-Head, eliminating the need for conventional NMS(Non-Maximum Suppression) post-processing. Our experiments on the GRDDC and IRRDD datasets demonstrate that CL-PSDD achieves competitive mean Average Precision (mAP) scores of 76% and 60%, respectively, rivaling the performance of supervised methods and excelling in generalization. This study pioneers the application of unsupervised learning to extensive pavement surface distress detection, establishing a significant milestone and providing a valuable reference for future research in this vital field.**

***Index Terms*— Pavement surface distress detection, unsupervised learning, contrastive learning, object detection.**

I. Introduction

ROADS are the bedrock of social infrastructure, essential for the establishment of efficient transportation systems. Yet they are perpetually subjected to environmental stressors such as rain, frost, and traffic pressure, leading to various forms of degradation. These deteriorations not only mar the road surface but also escalate the risk of traffic accidents. To counteract these issues, many nations have enacted extensive legal frameworks and stringent road quality standards [1] aimed at maintaining consistent adherence to maintenance and safety protocols. In this context, the timely detection of pavement surface distress is paramount to preserving road safety and ensuring the seamless functioning of transportation networks.

In recent years, the application of deep convolutional neural networks (DCNNs) has revolutionized the field of pavement surface distress identification. Scholars have harnessed the power of DCNNs to enhance feature encoding and detection techniques, leading to significant improvements in accuracy. Cha et al. [2] were pioneers in using convolutional neural networks (CNNs) for detecting road cracks. Arya et al. [3] further increased detection precision by introducing an ensemble model approach. Additionally, Guo and Zhang in [4] developed the MN-YOLOv5 model, which boosts detection efficacy for pavement surface distress through the adoption of a lightweight network structure, MobileNetv3. Despite the remarkable progress, DCNN-based methods face challenges due to their reliance on extensive, high-quality labeled datasets. Inconsistencies in labeling across datasets and the diversity of road conditions in different regions can impede a model's ability to generalize effectively. Addressing these challenges is critical for the development of robust and adaptable pavement surface distress detection systems.

Within the semi-supervised learning paradigm, traditional pavement surface distress detection methods have leveraged generative techniques to augment distress samples. Maeda et al. [5] introduced the PG-GAN model to expand the original dataset by creating new pavement surface distress samples through generative techniques. Similarly, Shim et al. [6] developed the SR-GAN model, which improves image resolution via generative models. Lin et al. [7] unveiled the DA-RDD model, which facilitates cross-domain adaptation for pavement surface distress detection through adversarial learning and enhances the original dataset with generative models. While these approaches have somewhat improved the model's generalization abilities, they often necessitate the development of complex network architectures to effectively manage domain differences between labeled and unlabeled datasets.

Unsupervised contrastive learning focuses on acquiring meaningful representations by highlighting differences between various samples. Its core principle is to maximize the similarity of samples within the same class while minimizing the similarity between samples of different classes. This approach ensures that the model learns representations that are similar within the same class but dissimilar across different classes. This concept was initially introduced by Hadsell et al. in [8]. Contrastive learning typically uses agent tasks for model training [9], [10], [11]. In [12], He et al. developed MoCo

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(Momentum Contrast), which uses a queue and momentum encoder to optimize the use of positive and negative samples for contrastive learning. Chen et al. introduced SimCLR (Simple Contrastive Learning) in [13], where they designed the projection head module that significantly boosts the model's performance. BYOL (Bootstrap Your Own Latent), proposed by Grill et al. in [14], eliminates the need for negative samples in contrastive learning by utilizing a prediction mechanism. Currently, research on pavement surface distress detection using unsupervised learning is limited. This paper aims to pioneer new methods for detecting pavement surface distress under the unsupervised learning paradigm.

End-to-end semantic segmentation models have demonstrated remarkable effectiveness in detecting pavement surface distress. For instance, Shim and Cho [15] proposed a lightweight model that leverages multi-scale analysis for road damage detection, while Liu et al. [16] developed an advanced UNet-based method tailored for complex environments. Despite their accuracy, these models often involve significant computational overhead due to their reliance on pixel-level recognition. In contrast, object detection techniques provide a more resource-efficient alternative that maintains accuracy and excels in identifying and localizing pavement surface distress. Currently, there are no available object detection models fully capable of end-to-end pavement surface distress detection. This paper explores the potential of employing end-to-end object detection models for practical applications in pavement surface distress detection.

In this paper, we present CL-PSDD (Contrastive Learning for Pavement Surface Distress Detection), a pioneering unsupervised learning model for pavement surface distress detection. CL-PSDD integrates SwAV and YOLO technologies to adeptly capture feature representations of complex pavement surface distress through unsupervised contrastive learning and clustering. Our experiments on the GRDDC [17] and IRRDD datasets, as illustrated in Fig.1, demonstrate that CL-PSDD achieves performance metrics on par with state-of-the-art supervised learning algorithms in terms of COCO AP. Moreover, it showcases superior generalization capabilities, highlighting the model's effectiveness and adaptability.

The main contributions of this paper are as follows:

(a) Proposal of the distress model: We introduce a groundbreaking unsupervised contrastive learning model for pavement surface distress detection in complex environments. Our approach innovatively combines SwAV with YOLO, leveraging their synergies to master the feature representations of pavement surface distress.

(b) Design of the RE-RPN-K region proposal module: We designed the RE-RPN-K module to extract road surface information and generate candidate boxes. This module enhances image texture detail through a series of transformations, effectively capturing pavement surface distress objects, reducing attention to other area interferences, and balancing training effectiveness with computational costs.

(c) Advantages of end-to-end target detection deployment: By redesigning the YOLO series detectors and incorporating the transformer-based DETR-Head, we removed the YOLO models' reliance on NMS post-processing. This advancement enables true end-to-end deployment of the CL-PSDD model, enhancing its practicality.

(d) Construction of the AU-RoadDataset-18K dataset: We have curated a large-scale unsupervised pavement surface distress detection dataset, AU-RoadDataset-18K, by compiling data from seven countries. Utilizing the RE-RPN-K algorithm, this dataset serves as an invaluable resource for future unsupervised pavement surface distress detection research. The dataset has been open-sourced and is available at: https://github.com/scorpio1121/AU-RoadDataset-18K.

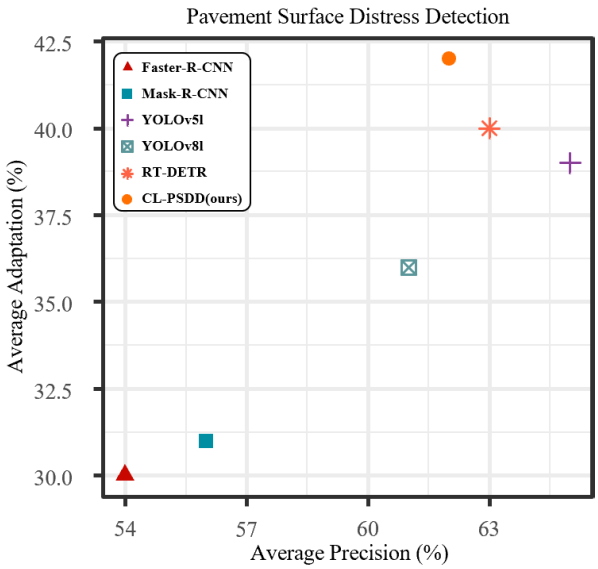

Fig. 1. Our CL-PSDD model, leveraging unsupervised learning, surpasses conventional supervised learning frameworks in pavement surface distress detection, demonstrating superior generalizability in Zero-Shot scenarios.

The paper is structured as follows: Section II provides a review of related work, Section III describes the implementation details of our proposed method, Section IV presents the relevant setup and results of the experiments, and Section V provides a summary of the paper.

II. Related Work

This section briefly reviews related work on pavement surface distress detection, unsupervised contrastive learning, and end-to-end target detection.

A. Pavement Surface Distress Detection

pavement surface distress detection techniques can be broadly classified into traditional methods and those based on deep learning. Traditional approaches typically employ shallow image features, utilizing algorithms such as the Sobel operator, Gabor filter, HOG (Histogram of Oriented Gradients) directional gradient, and LBP (Local Binary Pattern) [18], [19], [20], [21]. In contrast, deep learning-based methods

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leverage the robust characterization capabilities of deep convolutional neural networks to detect pavement surface distress. For example, Zhang et al. [22] developed a technique using CNNs with a sliding window mechanism to identify cracks on cement surfaces. Wan et al. [23] enhanced the YOLOv5 architecture with an ECA (Efficient Channel Attention) mechanism in their YOLO-LRDD model, aiming to improve detection accuracy. Further, Xiang et al. [24] introduced the YOLOv5s-DSG model to extract deeper semantic information from road scenes by deepening the network layers. Zheng et al. [25] incorporated contrastive learning and supervised adversarial learning into the LDBFSS (Latent Domain Background Feature Separation and Suppression) model, augmenting its discriminative capabilities with YOLOv5 technology. In summary, deep learning-based methods offer a more generalized and precise framework for detecting pavement surface distress. However, most existing research, including those mentioned, employs supervised or semi-supervised learning strategies. There is a noticeable gap in studies exploring unsupervised learning approaches within this field. This paper seeks to fill that gap by pioneering the application of unsupervised learning techniques to pavement surface distress detection, representing a novel stride in this research domain.

B. Unsupervised Contrast Learning

Contrastive learning, an unsupervised learning paradigm, aims to map high-dimensional input points to a low-dimensional space while ensuring that points similar in the input space are positioned closely in the mapped space. Caron et al. [26] SwAV, which employs class clustering to segment the source image into subgraphs. These subgraphs, representing different viewpoints, are then mapped into the clustering space to predict permutations. This method effectively reduces the distance between similar and dissimilar objects from the clustering centers. Chen and He [27] developed SimSiam, emphasizing two crucial steps to enhance the performance of self-supervised learning. Initially, an encoder network generates data representations, which are subsequently mapped to a simplified space via a projection network. Ultimately, a prediction network utilizes this simplified representation to predict the original representation, thereby driving encoder learning. Additionally, Caron et al. [28] with DINO and Chen et al. [29] with MoCoV3 explore the varying contributions of the Transformer architecture in contrastive learning. Pavement surface distress detection involves identifying a range of conditions such as cracks, potholes, depressions, and pavement deformations, which vary in shape, size, color, and texture. These varying characteristics demand a model capable of generalizing well and making accurate judgments. This paper introduces an unsupervised contrastive learning approach tailored for detecting pavement surface distress, enhancing the model's ability to discern subtle distinctions and similarities among the different forms of pavement surface distress.

C. End-to-End Target Detection

In the field of pavement surface distress detection, mainstream methods predominantly utiliz CNN-based techniques, such as Fast R-CNN [30], Faster R-CNN [7], [31], SSD [31], and YOLO [4], [23], [24], [25]. These methods typically rely on non-maximum suppression (NMS) for post-processing to eliminate overlap-ping detection boxes, a requirement that complicates achieving true end-to-end deployment on all hardware platforms during model deployment. To address this challenge, Carion et al. [32] introduced DETR, a groundbreaking target detection method that uses a transformer architecture and conceptualizes target detection as a set prediction problem. By employing the Hungarian algorithm for bipartite graph matching, DETR ensures that each ground truth box is optimally matched with one detection box, eliminating the need for NMS and facilitating genuine end-to-end deployment. This paper seeks to pioneer the integration of DETR with traditional CNN-based detection frameworks to reduce the dependency on NMS in pavement surface distress detection. Our goal is to advance the application of end-to-end target detection methods, which are inherently more efficient than end-to-end semantic segmentation, as they operate on image-level detection rather than pixel-level processing.

III. Methodology

This section delves into the implementation details of CL-PSDD (Contrastive Learning for Adaptive Generalized Pavement Surface Distress Detection), with the overall model structure depicted in Fig.2. CL-PSDD is a framework that combines unsupervised contrastive learning (SwAV) with single-stage object detection (YOLOv8) for road object detection. The framework comprises four main components: Proposals, SwAV, Future-Encoder, and DETR-Head. Proposals perform unsupervised comparative learning on the input road images with the SwAV module to update the feature extractor f_θ in the Future-Encoder module. The RE-RPN-K module is designed to achieve the regional proposal of road lesions. Meanwhile, to achieve end-to-end model deployment and address the YOLO framework's dependency on NMS post-processing, we introduce the transformer-based DETR-head detector. This detector predicts the bounding boxes and categories of targets based on the feature information obtained from the Future-Encoder module, thus realizing pavement surface distress detection.

A. Swapping Assignments and Pavement Surface Distress Detection

In the CL-PSDD framework, we introduce SwAV to learn the feature representation of images. With SwAV, the model can learn meaningful feature representations from multiple views of the same image, thereby enhancing the model's ability to detect complex pavement surface distress. The structural model of SwAV is illustrated in Fig.3. To distinguish supervised learning from unsupervised learning, formally we represent the labeled dataset as the set $M = \{(x_i^M, box_i^M, cls_i^M)_{i=1}^{N_M}\}$, where $x_i^M \in R^{w\times h}$ represents the source image, $box_i^M \in R^{n\times 4}$ represents the target's coordinate, $cls_i^M \in R^{n\times 1}$ represents the target's category and the number of samples X_i^M is denoted by N_M. The set of unlabeled data is denoted as $U = \{(x_i^U)_{i=1}^{N_U}\}$, where $x_i^U \in R^{w\times h}$ represents

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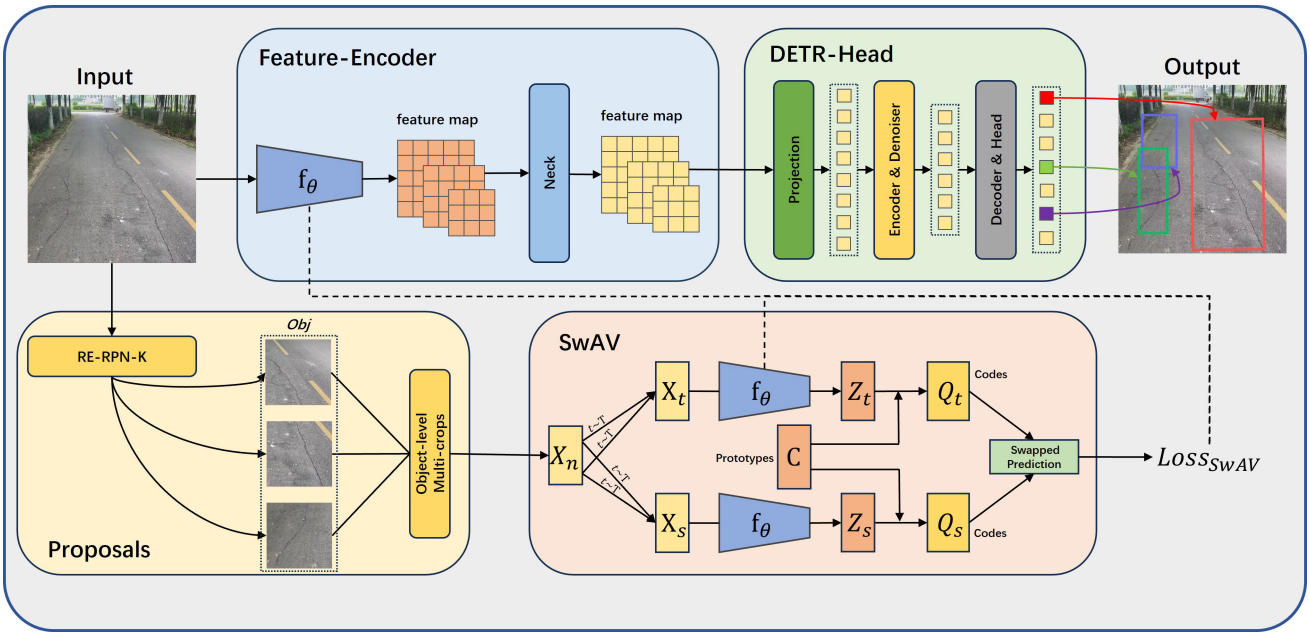

Fig. 2. The CL-PSDD network architecture involves obtaining K object-level images as inputs to the SwAV module through the region proposing RE-RPN-K module, for a road image. The SwAV module performs unsupervised contrast learning by comparing and clustering the images to update the feature extractor f_θ. The final feature map used for prediction is obtained by comparing f_θ with Neck using the Feature-Encoder. The DETR-head transforms the feature graph into a sequence of features using Projection. The encoder then takes this sequence as input and generates the target frame through object querying.

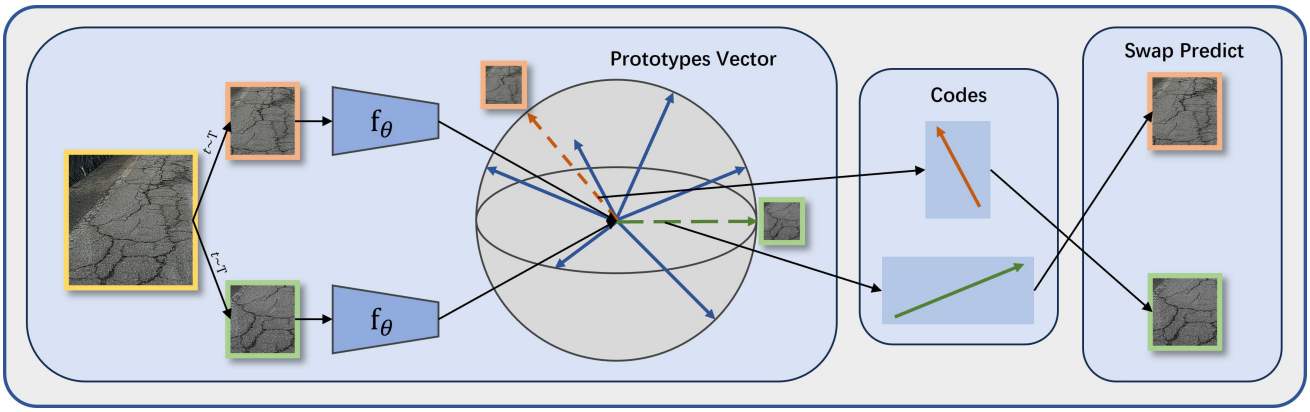

Fig. 3. The SwAV module allows for the mapping of feature vectors from a given input image in a spherical space. These feature vectors can predict each other even when they come from different viewpoint transformations of the same image, thanks to the acquired codes.

the source image, and x^U represents the number of samples x_i^U. The transform t is randomly sampled from the image transform set T for a given image X_n to obtain an enhanced image X_{nt}. The feature map is then obtained by the feature extractor $f_\theta$and projected into the unit sphere, resulting in $Z_{nt} = \frac{(f_\theta(X_{nt}))}{||(f_\theta(X_{nt}))||_2}$. The feature Z_{nt} is associated with a set of K learnable prototype vectors $C = \{(c_i^C)_{i=1}^K\}$, where $c_i^C \in R^{128\times 1}$. The feature code $Codes = Q_{nt}(Z_{nt}^\top \cdot C)$ is then computed, and the loss value of permutation prediction is calculated using the cross-entropy function. The loss function $Loss_{SwAV}$, shown in Mathematical Equation 1, is the sum

$$Loss_{SwCL} = -\left(\sum_k q_s^{(k)} \log\left(\frac{\exp\left(\frac{1}{\tau} Z_t^\top c_k\right)}{\sum_{k'} \exp\left(\frac{1}{\tau} Z_t^\top c_{k'}\right)}\right) + \sum_k q_t^{(k)} \log\left(\frac{\exp\left(\frac{1}{\tau} Z_s^\top c_k\right)}{\sum_{k'} \exp\left(\frac{1}{\tau} Z_s^\top c_{k'}\right)}\right)\right) \quad (1)$$

of the cross-entropy loss of predicting q_s by feature z_t and q_t by feature z_s. The temperature is denoted by τ. The loss function enables back-propagation to minimize both the prototype vector C and the parameter θ of the feature extractor f_θ, responsible for generating the features. Pavement surface distress detection requires a model with high generalization and judgment abilities due to the diversity and similarity of pavement surface distress. Our approach, as shown in Mathematical Equation 2, involves unsupervised comparative learning to maximize the probability of predicting similar distress and minimize the probability of predicting different.

$$\max Q((\frac{(f_\theta(t(X_i^U)))}{||(f_\theta(t(X_i^U)))||_2})^\tau C) \Longrightarrow ((\frac{(f_\theta(t(X_i^U)))}{||(f_\theta(t(X_i^U)))||_2})^\tau C),$$
$$\min Q((\frac{(f_\theta(t(X_i^U)))}{||(f_\theta(t(X_i^U)))||_2})^\tau C) \Longrightarrow ((\frac{(f_\theta(t(X_j^U)))}{||(f_\theta(t(X_j^U)))||_2})^\tau C),$$
$$where i, j \in N_U, i \neq j \quad (2)$$

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distress. This process aims to optimize the parameter θ of the feature extractor f_θ. This will enable it to differentiate complex pavement surface distress based on their similarity in real-life situations and improve the model's discriminative ability.

B. RE-RPN-K

Region proposal play a crucial role in unsupervised object detection. Unlike supervised datasets M containing target coordinates box_i^M and category information cls_i^M, unsupervised datasets U lack such annotations. Therefore, it's necessary to construct corresponding target coordinates box_i^U and category information cls_i^U from unsupervised datasets U. The Selective Search algorithm [33]is widely utilized for generating region proposals from image to object levels. However, in pavement surface distress detection, this algorithm may capture vehicles and pedestrians on the road surface, as well as trees and buildings along the roadside, often overlooking the pavement surface distress themselves. Pavement surface distress is typically not easily identifiable based on color, size, or scale. Nevertheless, when approached from a textural perspective, pavement surface distress exhibits a clear distinguishability. Therefore, we propose RE-RPN-K, aiming to enhance texture information, thereby enabling region proposals to focus more on road surface distress information. RE-RPN-K consists of the Road Enhanced Object Extractor (REOE) module and the space-K algorithm.

1) REOE: The pseudo-code for REOE is shown in Algorithm 1. Specifically, for a given image X_n, the enhanced image $X_{enhanced}$ is obtained by applying a series of image transformations T'. As shown in Mathematical Equation 3, Gaussian blurring is performed on the brightened gray map using a

$$X_{Ggray} = G * (brighten(X_n) \cdot [0.30, 0.58, 0.12]^\tau) \quad (3)$$

Gaussian kernel with a convolution kernel of 5 to obtain a Gaussian gray map X_{Ggray}, where brighten calculates the average luminance μ of X_n and performs a pixel increment operation when u is lower than a specific threshold before

Algorithm 1 Road Enhanced Object Extractor

```
# Input: Original image
# Output: A collection of bounding boxes
for x in loader:
    x_enchaned = aug(x) # texture enhancement
    I = to_binary(x_enhanced) # binarization processing
    Box = [ ] # initialization
    for pixel in I:
        If the pixel has an object and the pixel is not visited:

            visited(pixel) # make pixel visited
            row,col = g(pixel) # recursion around the pixel
and save coordinates
            Box.append((row.min,col.min,row.max-
row.min,col.max-col.min)) # x,y,w,h
    return Box
```

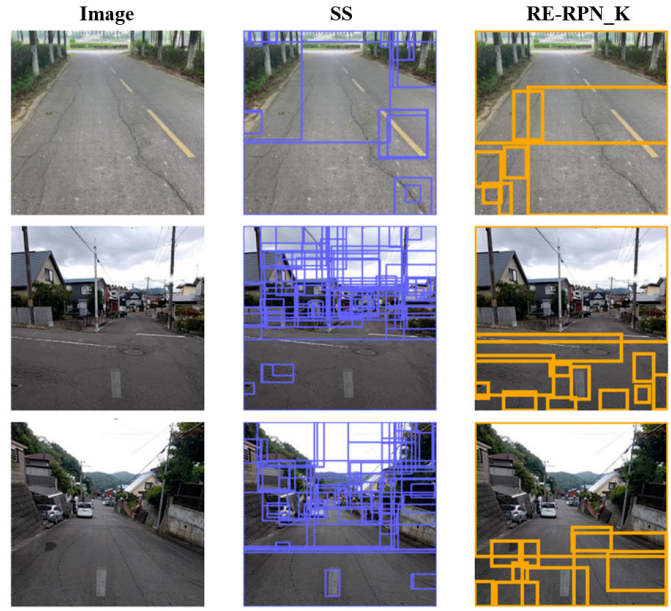

Fig. 4. This document presents the visual comparison results between the RE-RPN-K algorithm and the Selective Search algorithm.

pixel increment operation, in addition, the constant vector [0.30 0.58 0.12] is used to weight and sum the RGB three channels of the image to obtain the grayscale map. Then the Gaussian gray map X_{Ggray} is subjected to the Canny operator and morphological expansion iteration on horizontal gradient G_x and vertical gradient G_y to get the enhanced image $X_{enhanced}$ as shown in Mathematical Equation 4, where B is the morphological expansion and iter is the number of iterations, and we set iter to be 3. If the item is too small, it will lead to insufficient enhancement of texture information,

$$X_{enhanced} = \sqrt{G_x(X_{Ggray})^2 + G_y(X_{Ggray})^2} + B \times iter \quad (4)$$

and if iter is too tiny, iter will lead to insufficient enhancement of texture information. Finally, we retrieve the enhanced image $X_{enhanced}$ by biplot contour retriever g to get the pavement surface distress target set $L_{road} = \{(box_i^L)_{i=1}^{N_L}\}$, where $box_i^L \in R^{n\times 4}$ denotes the coordinates of the target, and N_L denotes the number of samples box_i^L. The difference between the effect of REOE and Selective Search on region proposal is shown in Fig.4. In addition, for different images, the number of important targets that exist is usually uncertain, the number of region proposals is usually large, and their order in the set is unclear.

2) Space-K: The dataset for pavement surface distress detection is typically captured by cameras mounted on vehicles. However, these images often contain irrelevant information, such as the sky and houses in the upper part of the picture. To address this issue, we developed the space-K algorithm. Using a two-dimensional spatial weighting algorithm, this algorithm assigns a score q_i^L to the candidate target box box_i^L. We designed two vectors $weight_x$, $weight_y \in \{0, 1\}^{1\times 10}$, to measure the scoring weights of the horizontal and vertical coordinates where

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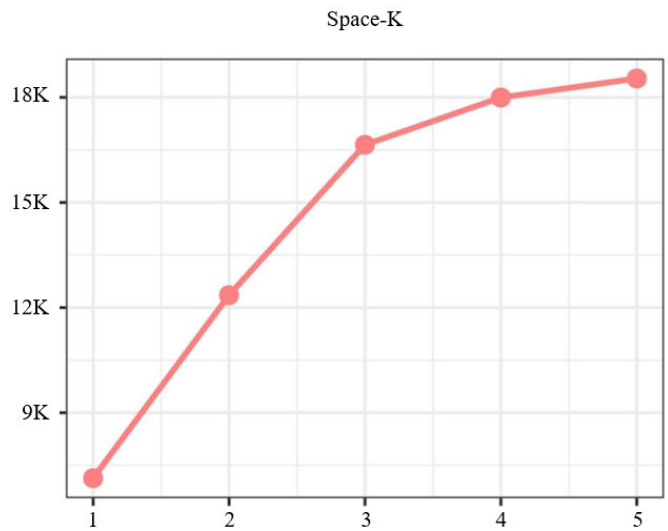

Fig. 5. The size of object-level datasets is affected by the K-values in the RE-RPN-K algorithm.

the center point of the target box is located. Expressly, we set $weight_x = [0, 0, 0, 1, 1, 1, 1, 1, 1, 1]$,$weight_y = [0,1,1,1,1,1,1,1,1,1,0]$. Mathematical Equation 5 shows that for each candidate box $box_i^L = \{x, y, w, h\}$, a score is calculated

$$Score(box_i^L) = x + \frac{w}{2} + weight_x[\frac{x}{w} + \frac{2}{2}](x + \frac{w}{2}) + y + \frac{h}{2} + weight_y[\frac{y}{h} + \frac{h}{2}](y + \frac{h}{2}) \quad (5)$$

and weighted based on the two-dimensional position of the picture and pre-set weights. This results in a set of pavement surface distress target boxes with the score of the target box $L_q = \{(box_i^L, q_i^L)\}_{i=1}^{N_L}$. Here, $box_i^L \in R^{n\times 4}$ denotes the target coordinates, q_i^L denotes the score of sample box_i^L, and N_L denotes the number of sample box_i^L. The target box set H_{road} is constructed by selecting the highest scores of the K target boxes in L_q, as shown in Mathematical Equation 6. Then, the dataset for unsupervised comparison learning is constructed as $U' = \{(x_i^{U'}, H_{road}{}_i^{U'})_{i=1}^{N_{U'}}\}$, where $x_i^{U'} \in R^{w\times h}$ denotes the source image and $H_{road}{}_i^{U'}$. The set of target frames in the

$$space - K(L_q, K) = argmax_{i\in(1,2,\ldots,N_L)} q_i^L, 1 \le K \le N_L \quad (6)$$

sample $x_i^{U'}$ for unsupervised training is denoted by $H_{road}{}_i^{U'}$ denotes the number of sample $x_i^{U'}$. Fig.5 shows the unsupervised dataset AU-RoadDataset-K constructed at different K values. The RE-RPN-K module enables the realization of a more applicable regional proposal for pavement surface distress detection.

C. DETR-Head

In target detection, post-processing is a commonly used method for removing redundant output frames. Mainstream CNN-based target detectors use NMS post-processing, such as Faster-R-CNN, YOLO, SSD, and Mask R-CNN. The set of prediction boxes, $L_{output} = \{(box_i^L, conf_i^L, cls_i^L)_{i=1}^{N_L}\}$, where $box_i^L \in R^{n\times 4}$ denotes the coordinates of the prediction box, $conf_i^L$ represents the confidence of the prediction box, while $cls_i^L \in R^{n\times 1}$represents the category of the target. N_L represents the number of prediction box samples box_i^L. In Mathematical Equation 7, the NMS sorts the set of prediction

$$\begin{aligned} L_{final} = \{(box_i^L, conf_i^L, cls_i^L)_{i=1}^{N_L} | conf_{\max}^L > conf_{threshold} \\ and \quad for i in N_L, cls_{maxconf}^L = cls_i^L \\ and \quad IoU(box_{maxconf}^L, box_i^L) < IoU_{threshold}\} \end{aligned} \quad (7)$$

boxes based on the box with the highest confidence level, $box_{maxconf}^L$, along with the other boxes of the same category. The IoU value is then calculated. When the Intersection over Union (IoU) value exceeds the pre-set threshold $IoU_{threshold}$, box_i^L will be suppressed, and $box_{maxconf}^L$ will be retained. Additionally, the confidence threshold $conf_{threshold}$ limits the number of final prediction boxes in L_{final}; it serves as the lower confidence limit in L_{final}. Modifying the hyperparameters $IoU_{threshold}$ and $conf_{threshold}$ can impact the model's performance. Additionally, the NMS operator requires specific hardware support. To realize the end-to-end deployment of pavement surface distress detection, we introduce the transformer-based DETR-head as a target detector. DETR-head transforms the target detection into an ensemble prediction problem by constructing a loss matrix $Cost_{matrix} \in R^{N_L\times N_{GT}}$ for dichotomous matching so that each target box box_i^{GT} in the actual box collection L_{output} has a unique optimal solution matching boxes in the predicted box collection L_{output}, where $cost_{i,j} = Loss_{target}(box_i^L, box_j^{GT}) + Loss_{cls}(box_i^L, box_j^{GT})$.

D. Loss Function

The CL-PSDD loss function consists of an unsupervised loss function and a detector loss function. The unsupervised comparative learning module uses $Loss_{SwAV}$ to measure the similarity between vectors using the vector inner product, and the prediction loss is measured using the cross-entropy loss function. The DETR-head detector uses a $Loss_{matrix}$, which adopts the Hungarian algorithm for optimal bisection matching of the loss matrix $Cost_{matrix}$, $Loss_{matrix} = \sum_{N_L\times N_{GT}} cost_{i,j}$. Mathematical Equation 8 shows the overall loss function of CL-PSDD.

$$Loss_{CL-PSDD} = \min(Loss_{SwCL}) + \min(Loss_{matrix}) \quad (8)$$

IV. Experiment and Discussion

This section details the experimental setup and evaluates the performance of the proposed unsupervised learning method to traditional supervised learning by highlighting differences in training configurations, dataset setups, and a range of performance metrics. We thoroughly explore various factors, including training settings, dataset specifics, comparisons between unsupervised and supervised learning, generalization performance, robustness testing, ablation studies, and further experiments to fine-tune specific components of our method. The results showcase the capabilities and advantages of our approach in addressing the challenges of pavement surface distress detection.

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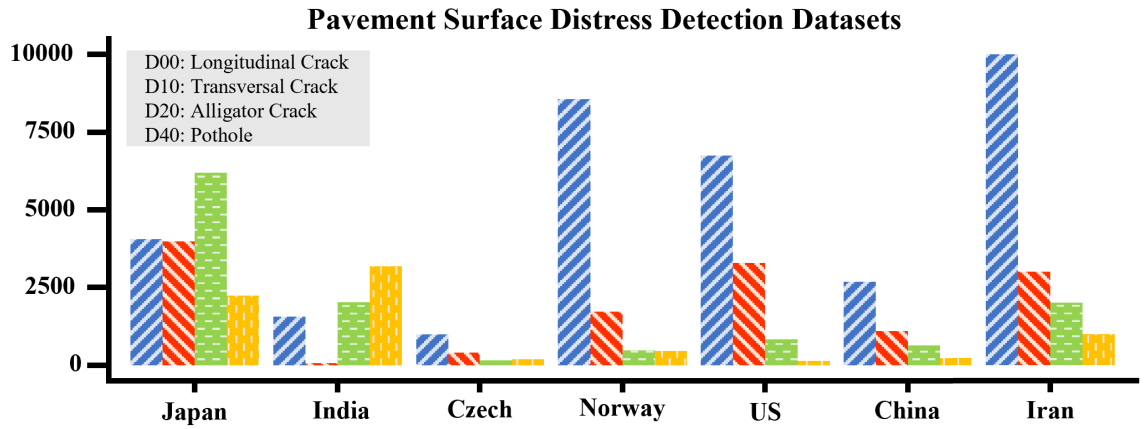

	Japan	India	Czech	Norway	US	China	Iran
D00	4096	1555	988	8570	6750	2678	9999
D10	3979	68	399	1730	3295	1096	3000
D20	6199	2021	161	468	834	293	2000
D30	2243	3187	197	461	135	86	1000

Fig. 6. GRDDC and IRRDD datasets: Details on Distress Categories.

A. Experiment Setting

The experimental code was implemented using the PyTorch framework was executed on the Ubuntu 18.04 operating system. We conducted the experiments using a Tesla V100 GPU (32GB). The batch size was set to 64, with training spanning 60 epochs for contrastive training and 100 epochs for supervised learning. We used the SGD optimizer with an initial learning rate of 0.01, a momentum of 0.937, and a weight decay of 5e-4. To ensure reproducibility of the results, the seed was fixed at 21.

B. Dataset Setting

We utilized two primary datasets for our experiments: GRDDC [17] and IRRDD, with detailed information presented in Fig.6. The GRDDC dataset includes pavement surface distress data from six countries: China, Japan, the United States, India, the Czech Republic, and Norway. It contains 47,420 images across four types of pavement surface distress: longitudinal crack (D00), lateral crack (D10), alligator crack (D20), and pothole (D40). The IRRDD dataset consists of 25,000 images of Iranian pavement surface distress, classified according to GRDDC criteria. Additionally, to create a dataset suitable for unsupervised learning, we also utilized the PPRDD and CNRDD [34]. PPRDD, proposed by Baidu Paddle during the 2021 Global Open Data Application Innovation Competition, contains 6,000 images of eight types of pavement surface distress: Crack, Manhole, Net, Pothole, Patch-Crack, Patch-Net, Patch-Pothole, and Other. CNRDD features pavement surface distress data from the G303 road section in China, including 4,319 images of eight distinct pavement surface distress types. We synthesized several datasets to construct the AU-RoadDataset-18K, which contains 185,931 object-level images at a resolution of 256 × 256, designed specifically for unsupervised learning. We synthesized several datasets to construct the AU-RoadDataset-18K, which contains 185,931 object-level images at a resolution of 256 × 256, designed specifically for unsupervised contrastive learning.

C. Results

1) Comparison With Classical Algorithms: We first trained the CL-PSDD model unsupervised on AU-RoadDataset-18K, followed by fine-tuning on labeled datasets to train target detectors. We compared its performance against mainstream supervised learning algorithms, including Faster R-CNN [35], Mask R-CNN [36], YOLOv5 [37], YOLOv8 [38], and RT-DETR [39]. The results, shown in Table I, indicate that CL-PSDD outperforms these supervised algorithms in terms of AP50 values across all three categories (D10, D20, and D40) of the IRRDD dataset. Although its performance on GRDDC is slightly lower than the highest recorded result, the CL-PSDD model demonstrates that our unsupervised contrastive learning approach is comparable to supervised learning methods, showcasing CL-PSDD's capability to learn feature representations in pavement surface distress through unsupervised comparison and clustering. Table I supports this conclusion. Additionally, the downscaled feature vectors derived from the model's Feature-Encoder were visualized using the T-SNE algorithm, as illustrated in Fig.7. The visualization reveals that the four distress categories form distinct clustering centers after deep clustering based on intrinsic similarity. To enhance clarity, We selected representative centers along with their associated samples. Samples with pronounced distress features cluster tightly around these centers, whereas those with less distinct features blend with other categories. Our analysis indicates that when there is a high degree of similarity and clear distinctions between categories, the clustering centers accurately reflect the intrinsic structure of the distress samples, residing in dense regions of the sample space. This results in homogeneous

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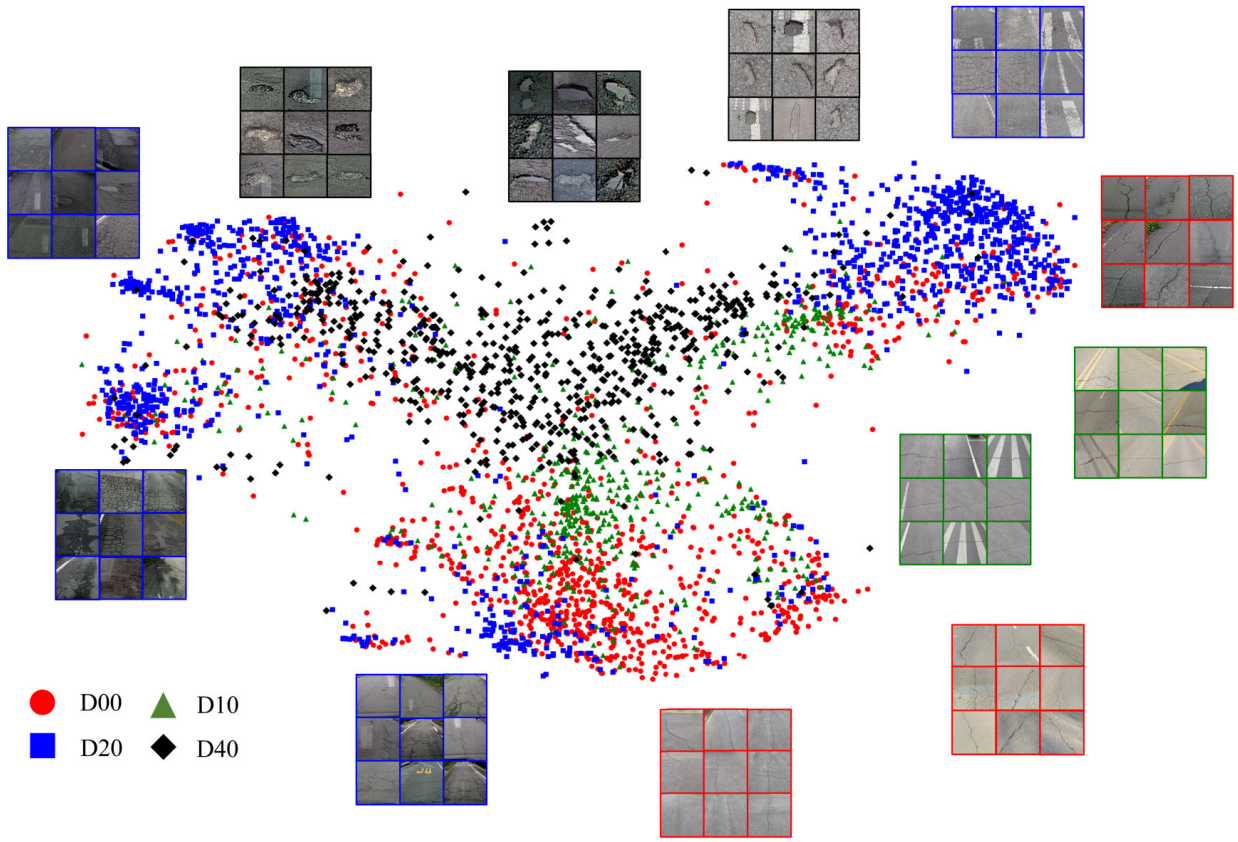

Fig. 7. Clustering visualization results of the feature extractor under Prototypes = 300.

TABLE I

THE EXPERIMENTAL RESULTS OF CL-PSDD VERSUS CLASSICAL ALGORITHMS WERE EVALUATED BASED ON THE AP VALUES AT IOU = 0.5 FOR FOUR DAMAGE CATEGORIES (D00, D10, D20, D40)

Method	IRRDD				GRDDC				GFLOPS
	D00	D10	D20	D40	D00	D10	D20	D40	
Faster-R-CNN [35]	0.703	0.642	0.795	0.384	0.560	0.503	0.669	0.436	134.4
Mask-R-CNN [36]	0.709	0.648	0.798	0.375	0.520	0.451	0.630	0.370	181.1
YOLOv5l [37]	**0.779**	0.742	0.844	0.650	0.604	0.596	0.691	0.517	135.3
YOLOv8l [38]	0.776	0.742	0.844	0.657	0.610	0.608	**0.712**	0.535	165.4
RT-DETR [39]	0.723	0.644	0.788	0.587	**0.612**	**0.609**	0.683	**0.544**	108.0
CL-PSDD(ours)	0.776	**0.753**	**0.851**	**0.662**	0.593	0.599	0.677	0.542	138.3

clusters that enhance the accuracy of distress identification and diagnosis. In contrast, samples with low similarity or minor differences yield less representative centers, often found at the sample space's edges or transitional areas. This leads to heterogeneous clusters, complicating classification and increasing diagnostic demands. Our findings highlight the importance of intrinsic similarity in classification and the limitations of current clustering methods for subtly different samples.

2) Generalizability Experiments: We tested the model on pavement surface distress datasets from multiple global regions to evaluate its generalization capabilities. Notably, we employed a ZeroShot evaluation method for cross-geographic assessment. The results, detailed in Tables II, III and IV demonstrate that our CL-PSDD model exhibits superior generalization performance. Fig.8 shows comparative result plots of mainstream target detection algorithms using supervised learning versus our CL-PSDD algorithm, which was trained using data from Japan and tested on datasets from various regions, including China, the United States, India, Norway, and the Czech Republic.

3) Robustness Experiments: We tested the robustness of our model under various noise conditions, with results detailed in Table V. We simulated real-world noise through Gaussian blur, occlusion, and random brightness adjustments. The results reveal that partial occlusion and mild brightness noise minimally affect the model's performance, causing only about

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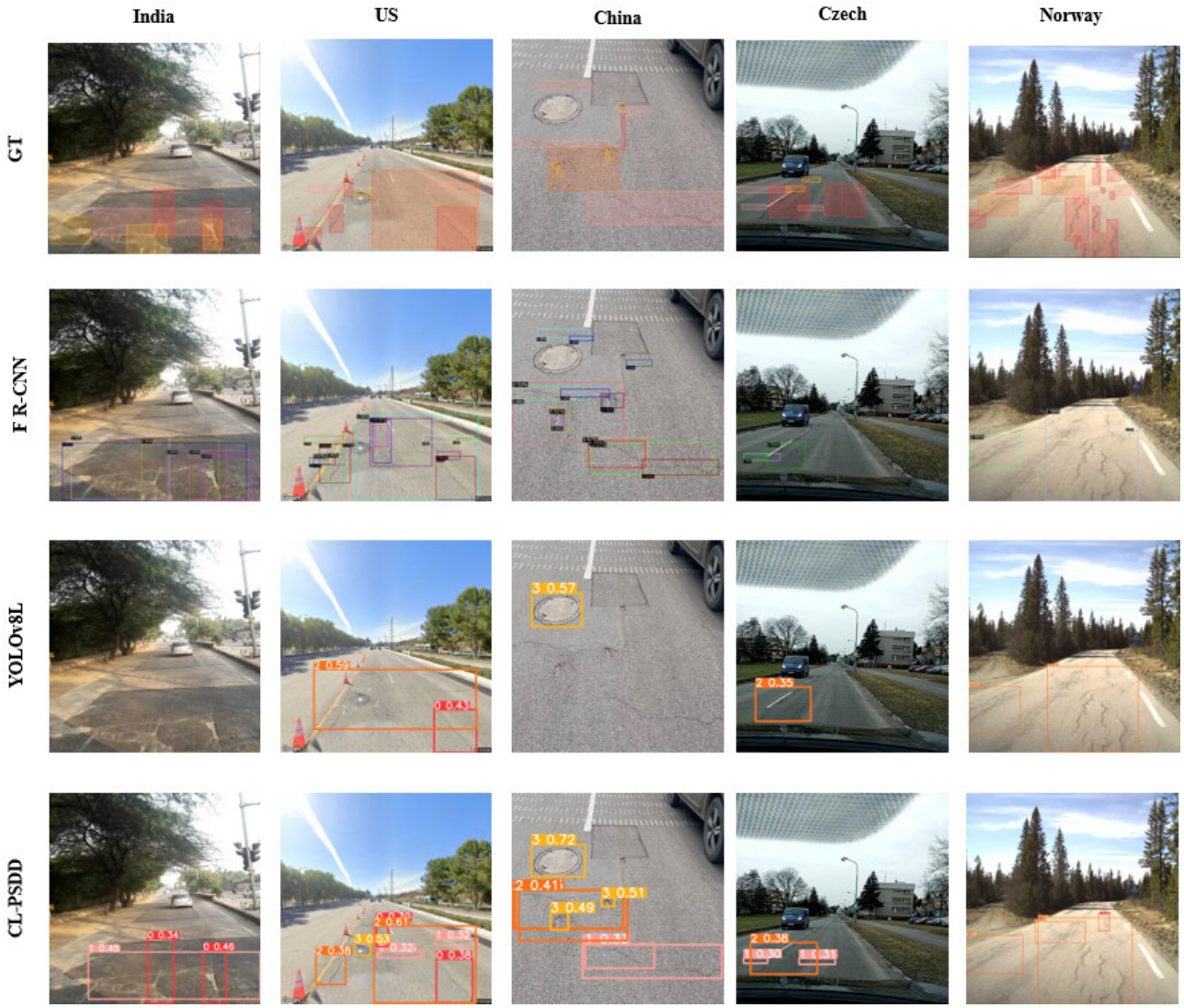

Fig. 8. Plots of the experimental results of Zero-Shot generalizability after training on the Japanese dataset.

TABLE II
GRDDC->IRRDD

Method	D00	D10	D20	D40
Faster-R-CNN [35]	0.44	0.387	0.541	0.062
Mask-R-CNN [36]	0.454	0.397	0.543	0.081
YOLOv5l [37]	0.377	0.335	0.48	0.091
YOLOv8l [38]	0.393	0.384	0.514	0.106
RT-DETR [39]	0.364	0.299	0.451	0.109
CL-PSDD(ours)	**0.554**	**0.524**	**0.583**	**0.229**

TABLE III
JAPAN -> CHINA

Method	D00	D10	D20	D40
Faster-R-CNN [35]	0.291	0.374	0.513	0.110
Mask-R-CNN [36]	0.284	0.325	0.492	0.134
YOLOv5l [37]	0.203	0.317	0.386	0.142
YOLOv8l [38]	0.219	0.33	0.431	0.144
RT-DETR [39]	0.240	0.364	0.44	0.110
CL-PSDD(ours)	**0.434**	**0.529**	**0.575**	**0.298**

a 5-10% variation. However, introducing strong noise, such as Gaussian blur, significantly degrades performance, with a 40-50% reduction. Gaussian blur consistently remains the main contributor to performance decline. Our observations indicate that supervised learning methods experience substantial performance losses under Gaussian blur due to their reliance on label information, which becomes less effective with strong noise in the input data. In contrast, our proposed CL-PSDD model, based on unsupervised contrastive learning, demonstrates remarkable robustness in high-noise environments. This resilience stems from CL-PSDD's ability to learn robust representations without relying on label information,

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TABLE IV
JAPAN -> US

Method	D00	D10	D20	D40
Faster-R-CNN [35]	0.549	0.357	0.537	0.233
Mask-R-CNN [36]	0.556	0.349	0.544	0.203
YOLOv5l [37]	0.572	0.443	0.446	0.075
YOLOv8l [38]	0.629	0.485	0.522	0.106
RT-DETR [39]	0.536	0.371	0.448	0.124
CL-PSDD(ours)	**0.624**	**0.573**	**0.625**	**0.319**

capturing essential structural features from extensive datasets. Enhanced through contrastive learning and clustering, the model's ability to discriminate among different features is significantly improved.

4) Ablation Experiments: To confirm the effectiveness of each component in our CL-PSDD model, we conducted ablation studies on the SwAV, RE-RPN-K, and DETR-Head modules. As shown in Table VI, introducing the SwAV module allowed for unsupervised training on unlabeled datasets, achieving average COCO AP50 values of 0.725 on the IRRDD dataset and 0.558 on the GRDDC dataset. The RE-RPN-K module, focusing on pavement information in regional proposals, improved the detection of pavement surface distress, increasing COCO AP50 to 0.746(+0.021) and 0.578(+0.02) on the IRRDD and GRDDC datasets, respectively. The transformer-based DETR-Head detector facilitated end-to-end target detection deployment, yielding COCO AP50 scores of 0.745 and 0.572 on the IRRDD and GRDDC datasets, respectively. Overall, the CL-PSDD showed improved scores of 0.760(+0.035) and 0.602(+0.044) on these datasets, validating the utility of each module. Further comparative experiments between RE-RPN-K and Selective Search revealed that RE-RPN-K significantly enhanced pavement surface distress detection, achieving higher COCO AP values of 0.760(+0.023) on the IRRDD dataset and 0.602(+0.039) on the GRDDC dataset, compared to those achieved with Selective Search, as shown in Table VII. These results underscore the advancements in detection capabilities with our model enhancements.

5) Exploring the Impact of Hyperparameters: We conducted an in-depth investigation into the influence of hyperparameters within our model's SwAV and RE-RPN-K modules, with a particular focus on their impact on the model's capacity to detect complex patterns and classify pavement surface distress more effectively. For the SwAV module, we experimented with the Prototypes' parameters across settings of 40, 100, 300, and 1000. As illustrated in Table VIII, incrementing the Prototypes from 40 to 300 notably improved the model's average COCO AP50 scores, rising from 0.721 to 0.760 on the IRRDD dataset and from 0.567 to 0.602 on the GRDDC dataset. These enhancements suggest that an expanded clustering space allows for a more nuanced feature representation, enabling the model to more adeptly capture complex target categories. However, increasing the Prototypes to 1000 did not yield additional improvements, indicating a performance plateau, likely due to the representational limitations of the unsupervised learning objectives and the finite variability within the dataset. Furthermore, we evaluated the K parameter in the RE-RPN-K module, with settings ranging from 2 to 5. The results, as detailed in Table IX, demonstrate that incrementally increasing K gradually enhances performance, culminating in peak COCO AP50 values of 0.760 and 0.602 on the IRRDD and GRDDC datasets, respectively, when K is set at 5. This indicates that a higher K value, by providing a greater number of training examples per iteration, bolsters the model's generalization across a wider array of data points, thereby enhancing its predictive accuracy for pavement surface distress detection. However, our observations suggest that beyond a certain threshold, increasing K minimally affects the provision of additional training examples and may inadvertently introduce irrelevant, texture-rich noise into the training set, potentially undermining the training process. These findings underscore the pivotal roles played by the dimensions of the clustering space and the number of samples per training batch in optimizing model performance, especially in complex unsupervised learning scenarios where discerning subtle inter-category differences is paramount.

6) Visualization: Fig.9 presents a t-SNE algorithm-based visualization of the feature vectors produced by our model's Feature-Encoder, highlighting the clustering performance across four distinct categories of pavement surface distress under varying prototype configurations. An increase in the prototype count refines the clustering centers' precision, leading to more pronounced and cohesive clusters. This heightened precision is essential for discerning the nuanced differences within pavement surface distress categories, thereby enhancing the model's diagnostic accuracy. Moreover, the denser clustering around the centers signifies a robust feature representation, which bolsters the model's ability to generalize across diverse pavement surface distress instances. Upon closer examination, we observed the effects of clustering with Prototypes set at 40, 100, and 300. With 40 Prototypes, initial clustering patterns were evident, yet the cluster boundaries were indistinct, and the samples' proximity to their respective centers was suboptimal. Elevating the Prototypes to 100 yielded clearer clustering centers and more definitive intercluster boundaries, with samples exhibiting a closer alignment to their centers. At 300 Prototypes, the clustering efficacy reached its zenith, with samples demonstrating significantly tighter alignment to their centers, thus maximizing the distinctiveness and unity of the clusters.

D. Limitations and Future Work

In this section, we summarize the limitations of the proposed CL-PSDD model and highlight potential directions for future research.

1) Limitations of Sample Resolution: Due to hardware constraints, the CL-PSDD model has primarily been trained and evaluated on low- to medium-resolution image datasets. As shown in Table X, we experimented with the Norwegian dataset at different resolutions, and the results demonstrate

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TABLE V

EXPERIMENTAL RESULTS ON ROBUSTNESS: THE VALIDATION SET WAS SIMULATED UNDER VARIOUS IMAGE TRANSFORMATIONS TO REPLICATE A REAL-WORLD APPLICATION ENVIRONMENT FOR ROBUSTNESS EVALUATION.PERFORMANCE WAS EVALUATED USING AP VALUES AT IoU = 0.5 FOR FOUR DAMAGE CATEGORIES (D00, D10, D20, D40)

Blur	Occasion	Bright	Method	IRRDD				GRDDC			
				D00	D10	D20	D40	D00	D10	D20	D40
			Faster-R-CNN [35]	0.353	0.218	0.531	0.278	0.296	0.234	0.356	0.264
			Mask-R-CNN [36]	0.365	0.231	0.502	0.269	0.264	0.209	0.311	0.236
✓			YOLOv5l [37]	0.331	0.381	0.52	0.229	0.29	0.228	0.344	0.192
			YOLOv8l [38]	0.339	0.397	0.513	0.249	0.292	0.256	0.387	0.159
			RT-DETR [39]	0.217	0.152	0.341	0.178	0.282	0.265	0.315	0.235
			CL-PSDD (ours)	**0.419**	**0.422**	**0.543**	**0.317**	**0.327**	**0.329**	**0.42**	**0.282**
			Faster-R-CNN [35]	0.626	0.587	0.748	0.36	0.506	0.453	0.636	0.409
			Mask-R-CNN [36]	0.613	0.596	0.756	0.357	0.498	0.436	0.574	0.322
	✓		YOLOv5l [37]	0.71	0.684	0.792	**0.623**	0.515	0.535	0.661	0.482
			YOLOv8l [38]	**0.714**	**0.687**	0.789	**0.623**	0.555	0.552	**0.689**	0.501
			RT-DETR [39]	0.696	0.681	**0.801**	0.605	**0.562**	**0.57**	0.671	**0.515**
			CL-PSDD (ours)	0.683	0.676	0.789	0.601	0.539	0.535	0.639	0.509
			Faster-R-CNN [35]	0.371	0.627	0.761	0.343	0.536	0.475	0.62	0.357
			Mask-R-CNN [36]	0.523	0.623	0.766	0.352	0.512	0.433	0.601	0.354
		✓	YOLOv5l [37]	0.776	0.735	0.841	0.64	0.595	0.584	0.676	0.505
			YOLOv8l [38]	**0.777**	0.736	0.84	**0.649**	0.603	0.605	**0.707**	**0.532**
			RT-DETR [39]	0.752	0.716	0.827	0.636	**0.611**	**0.607**	0.673	0.53
			CL-PSDD (ours)	0.765	**0.749**	**0.844**	0.641	0.571	0.573	0.649	0.514
			Faster-R-CNN [35]	0.308	0.183	0.465	0.247	0.262	0.205	0.312	0.241
			Mask-R-CNN [36]	0.297	0.192	0.377	0.243	0.235	0.183	0.278	0.216
✓	✓		YOLOv5l [37]	0.306	0.335	0.459	0.217	0.259	0.208	0.313	0.168
			YOLOv8l [38]	0.317	0.364	**0.479**	0.243	0.271	0.244	0.349	0.145
			RT-DETR [39]	0.184	0.117	0.297	0.154	0.253	0.253	0.267	0.209
			CL-PSDD (ours)	**0.371**	**0.371**	0.446	**0.253**	**0.283**	**0.287**	**0.357**	**0.242**
			Faster-R-CNN [35]	0.313	0.607	0.7	0.316	0.488	0.428	0.597	0.338
			Mask-R-CNN [36]	0.512	0.588	0.742	0.349	0.486	0.425	0.563	0.312
	✓	✓	YOLOv5l [37]	0.71	0.655	0.79	0.597	0.552	0.538	0.659	0.475
			YOLOv8l [38]	**0.716**	0.656	0.781	**0.609**	0.558	0.554	**0.685**	0.505
			RT-DETR [39]	0.696	**0.663**	**0.8**	0.59	**0.563**	**0.572**	0.661	**0.510**
			CL-PSDD (ours)	0.682	0.647	0.778	0.544	0.525	0.522	0.618	0.492
			Faster-R-CNN [35]	0.21	0.238	0.269	0.169	0.216	0.144	0.16	0.163
			Mask-R-CNN [36]	0.242	0.188	0.306	0.178	0.213	0.177	0.266	0.204
✓	✓	✓	YOLOv5l [37]	0.285	0.321	0.438	0.214	0.235	0.182	0.278	0.13
			YOLOv8l [38]	0.299	0.329	**0.446**	0.224	0.245	0.198	0.299	0.115
			RT-DETR [39]	0.175	0.125	0.263	0.143	0.228	0.218	0.246	0.162
			CL-PSDD (ours)	**0.349**	**0.351**	0.426	**0.231**	**0.256**	**0.259**	**0.304**	**0.219**

that higher resolutions significantly improve the model's performance. However, increasing resolution introduces several challenges, such as higher demands on hardware resources, difficulties in practical deployment, and a notable decrease in processing speed (FPS), which could hinder real-time applications of the algorithm. Future research should focus on overcoming these limitations to fully realize the model's potential for high-resolution image processing in real-world detection scenarios.

2) Challenges in Hyperparameter Optimization: The hyperparameters used in this studyincluding learning rate, batch size, and specific network architecture configurationswere optimized based on the characteristics of the GRDDC and IRRDD datasets. While these hyperparameters have

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TABLE VI
CL-PSDD ABLATION EXPERIMENT RESULTS: PERFORMANCE IS EVALUATED BY THE AP VALUE OF 4 DAMAGE CATEGORIES (D00, D10, D20, D40) AT IoU = 0.5

SwAV	RE-RPN-K	DETR-Head	IRRDD				GRDDC			
			D00	D10	D20	D40	D00	D10	D20	D40
✓			0.758	0.718	0.81	0.617	0.55	0.557	0.641	0.487
✓	✓		0.762	0.739	0.836	0.649	0.572	0.579	0.657	0.506
		✓	0.749	0.72	0.815	0.632	0.566	0.561	0.649	0.515
✓	✓	✓	**0.776**	**0.753**	**0.851**	**0.662**	**0.593**	**0.599**	**0.677**	**0.542**

TABLE VII
PROPOSAL ABLATION EXPERIMENT RESULTS: PERFORMANCE IS EVALUATED BY THE AP VALUE OF 4 DAMAGE CATEGORIES (D00, D10, D20, D40) AT IoU = 0.5

Proposals	IRRDD				GRDDC			
	D00	D10	D20	D40	D00	D10	D20	D40
SS [32]	0.759	0.732	0.821	0.637	0.556	0.563	0.637	0.499
RE-RPN-K	**0.776**	**0.753**	**0.851**	**0.662**	**0.593**	**0.599**	**0.677**	**0.542**

TABLE VIII
PROTOTYPES ABLATION EXPERIMENT RESULTS: PERFORMANCE IS EVALUATED BY THE AP VALUE OF 4 DAMAGE CATEGORIES (D00, D10, D20, D40) AT IoU = 0.5

Prototypes	IRRDD				GRDDC			
	D00	D10	D20	D40	D00	D10	D20	D40
40	0.748	0.711	0.816	0.611	0.558	0.557	0.635	0.519
100	0.764	0.729	0.83	0.627	0.576	0.578	0.653	0.521
300	0.776	**0.753**	**0.851**	0.662	**0.593**	0.599	**0.677**	0.542
1000	**0.778**	0.752	0.847	**0.663**	0.592	**0.603**	0.676	**0.544**

TABLE IX
SPACE-K ABLATION EXPERIMENT RESULTS: PERFORMANCE IS EVALUATED BY THE AP VALUE OF 4 DAMAGE CATEGORIES (D00, D10, D20, D40) AT IoU = 0.5

space-K	IRRDD				GRDDC			
	D00	D10	D20	D40	D00	D10	D20	D40
K=2	0.757	0.729	0.82	0.633	0.568	0.567	0.648	0.523
K=3	0.769	0.735	0.834	0.647	0.58	0.585	0.656	0.531
K=5	**0.776**	**0.753**	**0.851**	**0.662**	**0.593**	**0.599**	**0.677**	**0.542**

demonstrated generalizability within the field of road surface damage detection, their applicability in other domains remains untested. Future studies could explore automated hyperparameter tuning and meta-learning techniques to improve the model's adaptability and performance across a wider variety of datasets.

3) Sensitivity to Camera Angles: Most of the training samples used in this study were captured at angles between 30° and 45° relative to the ground plane. As a result, the model has not yet effectively learned representations for samples captured from other angles, such as 90° top-down views. When evaluating the GRDDC dataset, which includes aerial images captured by drones, the model's detection performance dropped significantly, as shown in Table XI. This performance decline may be attributed to severe distortion or occlusion of damage patterns from different angles, leading to significant shifts in data distribution and making it harder for the model to accurately detect damage features. Future research should explore angle-invariant feature extraction strategies and incorporate multi-angle training data to enhance the model's robustness. By acknowledging these limitations and providing insights into possible research directions, we aim to encourage

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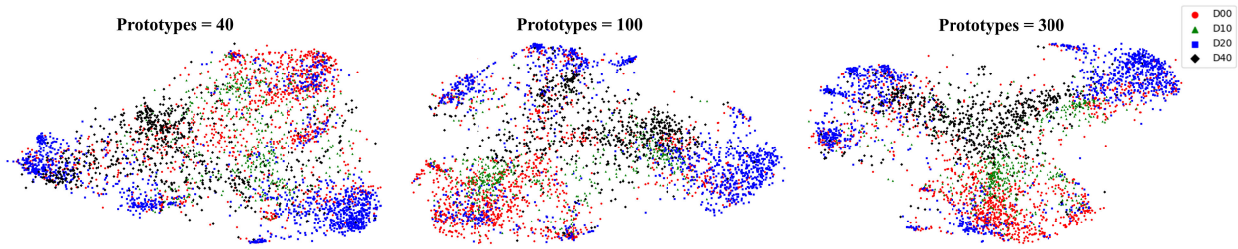

Fig. 9. Clustering visualization results of the feature extractor under Prototypes = [40,100,300].

TABLE X
NORWAY REGION OF THE GRDDC DATASET: 128 × 128, 256 × 256 AND 1024 × 1024 RESOLUTION

Resolution	Norway				FPS
	D00	D10	D20	D40	
128 × 128	0.564	0.566	0.623	0.550	103.1
256 × 256	0.588	0.578	**0.648**	0.578	84.0
1024 × 1024	**0.594**	**0.583**	0.643	**0.580**	14.7

TABLE XI
TEST RESULTS OF 30-45° AND 90° VIEWING ANGLE SAMPLES

View Angle	GRDDC			
	D00	D10	D20	D40
30-45°	0.593	0.599	0.677	0.542
90°	0.316	0.307	0.352	0.287

further advancements in the field of pavement surface distress detection, ultimately contributing to the development of more effective and scalable solutions.

V. CONCLUSION

This research introduces CL-PSDD, a novel contrastive learning-based approach for pavement surface distress detection, demonstrating remarkable generalization and robustness across extensive datasets. The CL-PSDD leverages SwAV for unsupervised contrastive learning, adeptly capturing intricate pavement surface distress features through comparative analysis and clustering. We have also introduced the RE-RPN-K module to accentuate texture information on road surfaces, thereby bolstering crack detection capabilities, particularly within texture-based candidate regions. Moreover, we have re-engineered the YOLO Head with a transformer-based DETR-Head, eliminating the necessity for NMS post-processing by harnessing the global modeling prowess of transformers and the Hungarian algorithm for optimal object matching. Our extensive AU-RoadDataset-18K underpins the validation of this method, with experimental results on the IRRDD and GRDDC datasets achieving average COCO AP50 metrics of 0.760 and 0.602, respectively. These results indicate a superior generalization capability compared to supervised methods. This study sets a precedent for further progress in unsupervised pavement surface distress detection, potentially enriching deep learning applications within the transportation sector.

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